

Semester Project Assignment — Agentic AI & RAG Systems

Course: Agentic AI

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1. Project Overview

For your semester project, you are free to choose any topic related to:

- **Agentic AI**
- **Retrieval-Augmented Generation (RAG)**
- **Agentic RAG Systems**
- **Improvements in agent orchestration, tool use, memory, reasoning, or actions**

Your project may follow **one of two tracks**:

TRACK A — Practical Agent System

Build a **fully functional agent** that performs **real actions**, not just text generation.

Examples of acceptable agentic behaviors

- An agent that reads data from a website and updates a Google Sheet
- An agent that analyzes medical text and flags risk factors
- An autonomous research agent that searches the web using APIs and summarizes findings
- A customer service agent that interacts with a database and updates entries
- A file automation agent (creates folders, edits files, classifies documents)
- A Slack/Discord bot that takes actions, sends reminders, or schedules tasks
- An AI tutor agent that retrieves content and generates structured lessons
- A programming assistant agent that runs code, tests, and reports errors
- A personal assistant that uses calendars, emails, and reminders via APIs

Important:

Your agent **MUST** be connected to a tool or action system such as:

- LangChain Tools
- OpenAI / Anthropic Function Calling
- Graphene / HumanEval / AgentEval
- HuggingFace Tools
- Custom tools/API integration

You MUST evaluate your agent using **any agent evaluation platform**, such as:

- **Graphene (OpenAI Agent Benchmarking)**
 - **AgentEval**
 - **ReActBench**
 - **HumanEval**
 - **AgentBoard tasks**
 - **Custom task evaluation** (if justified)
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TRACK B — Research Paper on Agentic AI / RAG

Write a **research paper** proposing an improvement in:

- Agent planning / reasoning
- Tool calling strategies
- Memory management
- Retrieval pipeline (RAG)
- Hybrid Agentic RAG
- Model prompting methods
- Multi-agent collaboration
- Evaluation frameworks for agents

Your paper must:

- Implement baseline models
- Propose a modification or new idea

- Compare performance with SOTA or classical methods
- Use proper experimental design
- Include clear evaluation metrics

Allowed models/tools for experiments:

- GPT models (OpenAI)
 - Claude models
 - Gemini
 - Llama / Mistral (via HF)
 - LangChain, LlamaIndex
 - ColBERT, DPR, FAISS, ChromaDB etc.
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2. Project Requirements

A. For Practical Agent Systems

Your submission must include:

1. System Architecture Diagram

- Show agent, tools, memory, RAG components, etc.

2. Description of the Agent

- Tasks performed
- Tools used
- Reasoning style (ReAct, MRKL, Planner-Executor, etc.)
- Technologies used (LangChain, OpenAI Functions, etc.)

3. Implementation

- Code repository (GitHub required)
- Documentation
- Environment setup instructions

4. Agent Evaluation

You must evaluate your agent using at least **5** benchmarking method:

Evaluation platforms & metrics may include:

- Graphene agent task accuracy
- Task success rate
- Action execution correctness
- Tool usage correctness
- Latency
- Reasoning length
- Cost efficiency
- Retrieval quality (for RAG-based agents)

5. Demo Video (3–7 mins)

- Show the agent performing actions
- Show evaluation results
- Clear explanation

B. For Research Paper Track

Your research paper must be **6–10 pages (IEEE format)** and include:

1. **Abstract**
2. **Introduction (problem definition + motivation)**
3. **Related Work**
4. **Methodology & Proposed Idea**
5. **Models Used**
6. **Experiments & Evaluation**
7. **Comparison With Previous Techniques**
8. **Discussion**
9. **Conclusion & Future Work**

10. References

Your experiments MUST include:

- At least **2 baseline models**
 - At least **1 proposed improvement**
 - Evaluation using at least **three metrics**, e.g.:
 - Accuracy / F1
 - Retrieval Precision@K
 - Hallucination rate
 - Agent success rate
 - Latency / throughput
 - Cost reduction
 - Relevance scores
 - Task completion time
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3. What You Will Be Evaluated On

Both tracks will be graded using the following rubric:

A. Model Quality (25%)

- Choice of model (GPT-4.1, Llama 3.1, Mistral, Claude, etc.)
- Complexity of architecture
- Justification of model choices

B. Agent Capabilities / Technical Work (25%)

- Ability to perform actions (not just text)
- Integration with tools/APIs
- RAG/Memory use (if applicable)
- The quality and novelty of your idea

C. Evaluation & Metrics (20%)

- Use of an evaluation platform (Graphene preferred)
- Clear evaluation metrics
- Interpretation of results
- Comparison with baselines

D. Research Quality & Completeness (20%)

- Structured write-up
- Depth of analysis
- Comparison with previous techniques
- Literature review quality

E. Presentation & Demo (10%)

- Clear demonstration
- Good explanation
- Professional delivery

4. Deliverables

For both tracks:

1. **GitHub Repository** (well-organized code)
2. **README Documentation**
3. **Evaluation Report**
4. **Final Research Paper (PDF)** — Optional for practical track, mandatory for research track